

RMIT

ISYS3476 (Postgraduate)
Managing Semi-structured and Unstructured Data
Assignment 1 Part B: Search Engine Implementation
Rubric

1 At-a-glance marking summary

Component	Points
Task 1: Indexing and structured retrieval	8
Task 2: Ranking	8
Task 3: System optimisation + MAP	9
Recorded CLI demonstration and technical explanation	3
Whole-submission code quality	2
Total	30

2 Marking policies and common rules

Pass rate. The fraction of hidden cases answered exactly correctly for a subtask.

Runtime. A 5-minute wall-clock safety cutoff applies per task.

Required interfaces. We do not adapt student code during marking. If a changed required contract prevents grading, the affected component receives 0.

Reporting evidence. Your submitted code and required run files must reproduce the evidence used for each reporting component. If the reproduced results do not match those reported in `WRITEUP.pdf`, the affected reporting component receives 0. An isolated transcription error does not trigger this rule. Unsupported evidence is disregarded, and the remaining evidence determines the band. Only the first page of `WRITEUP.pdf` is marked.

3 Task rubrics

3.1 Task 1 (8 points): Indexing and Structured Retrieval

Index package (2 points).

Component	Pts	Assessed behaviour
Wildcard index	1	Correct term sets per character n -gram, sorted and deduplicated outputs, and exclusion of unigram "\$".
Proximity index	1	Correct unigram position lists, sorted and deduplicated outputs, and [] for OOV terms.

Index scoring. Each index component is scored separately: 1 point if all hidden checks pass; otherwise 0 points.

Structured query processing (6 points). Hidden queries require exact document ID sets.

Processor	Pts	Assessed behaviour
Boolean	2	Boolean queries described in the Main Specification.
Wildcard	2	Wildcard queries described in the Main Specification.
Proximity	2	Proximity queries described in the Main Specification.

Query scoring. Each processor is scored separately: 2 points for pass rate = 1.00; 1 point for $0.50 \leq \text{pass rate} < 1.00$; 0 points for pass rate < 0.50.

Efficiency cap. A failed hidden stress test caps the affected query processor at **1 point**.

3.2 Task 2 (8 points): Ranking

Performance (4 pts): hidden test Pearson correlation.

Efficiency/stress-test cap. A failed hidden efficiency/stress test caps this component at **3 points**.

Points	Hidden Pearson correlation band
4	$\rho \geq 0.570$.
3	$0.540 \leq \rho < 0.570$.
2	$0.450 \leq \rho < 0.540$.
1	$0.200 \leq \rho < 0.450$.
0	$\rho < 0.200$.

Development & reporting (4 pts): in WRITEUP.pdf.

Pts	Evidence expected in write-up (first page only)
4	Meets the 3-point standard. A controlled comparison between default and an additional setting is grounded in concrete query and document or term evidence. The analysis clearly and convincingly explains how the setting changes the ranking mechanism and document scores, then explains the Pearson result in light of how Pearson correlation is calculated and what it measures about graded relevance. A clear evidence-led development trajectory weighs stability, limitations, and trade-offs and supports adopting or rejecting the setting.
3	Compared methods, including any combined or modified design, are clearly and sufficiently described. Reproducible dev evidence supports a sound development trajectory and judgement across queries or settings. The analysis gives a sound causal interpretation of how the ranking algorithm's scoring principles produced the observed score or ranking behaviour and Pearson outcomes. This interpretation is convincingly demonstrated by the cited evidence, but lacks the depth or precision of the highest band.
2	The method is identifiable and the results broadly reproducible, but the description is limited. Concrete dev results and a plausible explanation are present, but the causal interpretation is asserted rather than convincingly demonstrated by cited evidence.
1	Superficial, unclear, or incomplete method description, evidence, or judgement. Claims are difficult to connect to the submitted code.
0	Missing, not meaningful, or materially inconsistent with the submitted ranking code.

3.3 Task 3 (9 points): System Optimisation + MAP

Performance (4 pts): MAP on hidden test. Hidden MAP evaluates the explicit **default** output using its first 10 document IDs per query.

Points	Hidden MAP band
4	$\text{MAP} \geq 0.34$.
3	$0.32 \leq \text{MAP} < 0.34$.
2	$0.20 \leq \text{MAP} < 0.32$.
1	$0.05 \leq \text{MAP} < 0.20$.
0	$\text{MAP} < 0.05$, not runnable end-to-end, or invalid run output.

Efficiency/stress-test cap. A failed hidden pipeline-efficiency test caps this component at **3 points**.

Development & reporting (5 pts): in WRITEUP.pdf.

Pts	Evidence expected
5	The optimisations and any combined final system are clearly and sufficiently described. Two distinct optimisations are isolated in reproducible runs. A coherent and convincing analysis uses concrete query and document evidence to link their information retrieval mechanisms and interactions to ranking changes, then explains why MAP differs in light of how it is calculated. A clear evidence-led development trajectory demonstrates strong system-level judgement, weighs trade-offs, and supports the final choice and expected generalisation.
4	One isolated optimisation and its role are clearly described using concrete query and document evidence. A sound evidence-led analysis convincingly demonstrates how the optimisation's underlying information retrieval mechanism produced the ranking changes and resulting MAP outcome, in light of how MAP is calculated, and supports the final-system choice. This band includes reports evaluating only one well-supported optimisation.
3	The pipeline changes and dev results are identifiable and reproducible. Concrete query or document results and a plausible explanation are present, but the causal interpretation is asserted rather than convincingly demonstrated, and the system-level interpretation remains limited.
2	A broadly reproducible run comparison is present and the optimisation partly described, but the analysis remains descriptive, weakly connected to ranking and MAP, or difficult to verify.
1	Minimal or superficial. Evidence, optimisation description, or interpretation is unclear, incomplete, inconsistent, or generic.
0	Missing, not meaningful, no reproducible optimisation run, or materially inconsistent with the submitted code and run files.

3.4 Recorded CLI demonstration and technical explanation (3 points)

Points	Criteria
3	Excellent technical demonstration. The required live workflow is completed successfully. A reproducible Pearson or MAP result and the relevant submitted code are shown. The explanation gives a complete and accurate causal account of how the shown implementation affects document scores or rankings and how that behaviour produces the displayed result in terms of how Pearson correlation or MAP is calculated.
2	Sound live demonstration. Commands and navigation succeed, and the relevant code and discussed result are shown. A relevant causal explanation is present, but remains incomplete or imprecise in connecting the implementation, resulting score or ranking behaviour, and displayed metric.
1	Limited live demonstration. It runs, but mainly restates code or output, discusses an unseen result, or provides no meaningful causal explanation of the retrieval behaviour. Also applies when the demonstration is fragmented by repeated unsuccessful attempts or unrelated switching, only pre-generated output is shown, or major errors obstruct the demonstration.
0	Missing or not assessable. The recording is missing or unplayable, visibly edited, submitted only as an external link, fails to demonstrate the submitted system, or is materially inconsistent with the submitted files.

Whole-submission code quality (2 points)

Points	Criteria
2	Clear, consistent, and maintainable code with useful documentation, appropriate reuse, and defensive handling. No obvious dead code or unused imports. Required interfaces, layout, and the policy file are preserved.
1	Runnable and readable code with basic documentation and intact required files, but with minor duplication, style, unused-import, or defensive-handling issues.
0	Difficult to follow, largely undocumented, frequently crashes, uses disallowed libraries, has inappropriate side effects, breaks required interfaces or layout, makes no meaningful use of the student's Part A work, or alters the policy file.